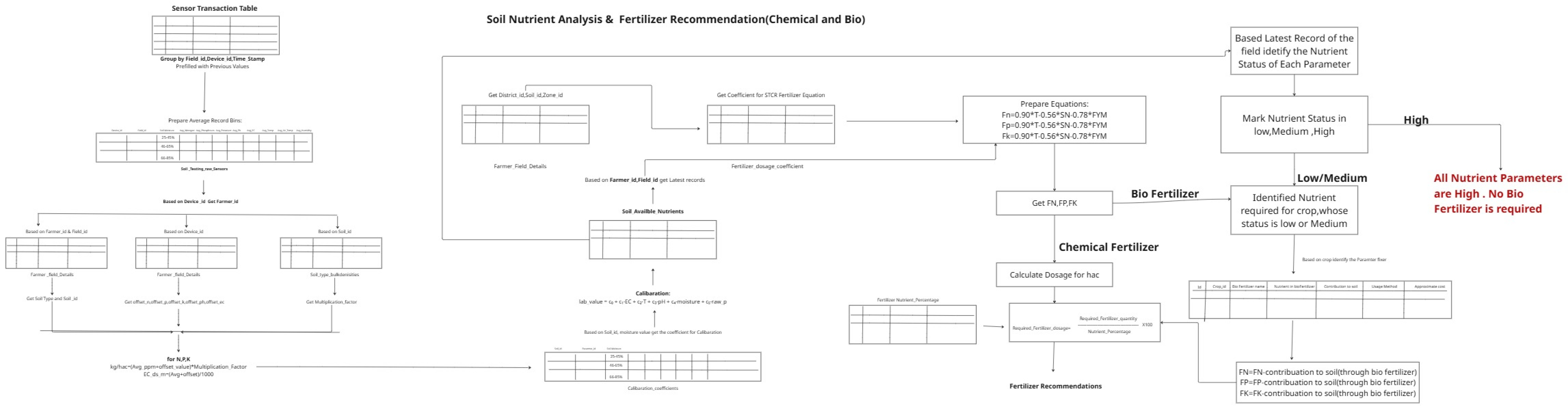
Soil Nutrient Analysis & Fertilizer Recommendation(Chemical and Bio)


Sensor Transaction Table
Group by Field_id,Device_id,Time_Stamp
Prefilled with Previous Values
Prepare Average Record Bins:
Device_id
Field_id
Soil Moisture
Avg_Nitrogen
Avg_Phosphorous
Avg_Potassium
Avg_Ph
Avg_EC
Avg_Temp
Avg_Air_Temp
Avg_Humidity
25-45%
46-65%
66-85%
Soil _Testing_raw_Sensors
Based on Device _id Get Farmer_id
Based on Farmer_id & Field_id
Based on Device_id
Based on Soil_id
Farmer _field_Details
Farmer _field_Details
Soil_type_bulkdenisities
Get Soil Type and Soil _id
Get offset_n,offset_p,offset_k,offset_ph,offset_ec
Get Multiplication_factor
for N,P,K
kg/hac=(Avg_ppm+offset_value)*Multiplication_Factor
EC_ds_m=(Avg+offset)/1000
Soil_id
Parameter_id
Soil Moisture
25-45%
46-65%
66-85%
Calibaration_coefficients
Based on Soil_id, moisture value get the coefficient for Calibaration
Calibaration:
lab_value = $c_0 + c_1 \cdot EC + c_2 \cdot T + c_3 \cdot pH + c_4 \cdot moisture + c_5 \cdot raw\_p$
Soil_Availble_Nutrients
Based on Farmer_id,Field_id get Latest records
Get District_id,Soil_id,Zone_id
Farmer_Field_Details
Get Coefficient for STCR Fertilizer Equation
Fertilizer_dosage_coefficient
Prepare Equations:
Fn=0.90*T-0.56*SN-0.78*FYM
Fp=0.90*T-0.56*SN-0.78*FYM
Fk=0.90*T-0.56*SN-0.78*FYM
Get FN,FP,FK
Bio Fertilizer
Chemical Fertilizer
Calculate Dosage for hac
Fertilizer Nutrient_Percentage
Required_Fertilizer_dosage= Required_Fertilizer_quantity / Nutrient_Percentage X100
Fertilizer Recommendations
Based Latest Record of the field idetify the Nutrient Status of Each Parameter
Mark Nutrient Status in low,Medium ,High
High
All Nutrient Parameters are High . No Bio Fertilizer is required
Low/Medium
Identified Nutrient required for crop,whose status is low or Medium
Based on crop identify the Paramter fixer
Id
Crop_id
Bio Fertilizer name
Nutrient in bioFertilizer
Contribution to soil
Usage Method
Approximate cost
FN=FN-contribuation to soil(through bio fertilizer)
FP=FP-contribuation to soil(through bio fertilizer)
FK=FK-contribuation to soil(through bio fertilizer)